

PARISIMA ABDALI

New York, NY

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Research Statement

I am passionate about using ML and DL models to tackle complex problems. One domain which I was always curious about is DL application in healthcare. With a strong background in electrical engineering and expertise in signal and image processing, I have worked on various projects, including my master's thesis on AI-generated MR images. Moving forward, I aim to continue advancing DL applications and contribute innovative solutions to science and technology.

Education

New York University, Tandon School of Engineering

New York, USA

Masters of Science in Electrical Engineering

Fall. 2022 - May. 2024

Thesis: MR Contrast Image Synthesis Using Deep Learning under Prof. Yao Wang

Affiliated with: NYU Video Lab, Rapid Imaging Lab

Imam Khomeini International University

Qazvin, Iran

Bachelor of Science in Electrical Engineering; (Ranked 1st)

Fall. 2016 – Summer. 2020

Research Interest

- Image and Video Processing
- Machine Learning
- Deep Learning
- Biomedical Signal Processing
- Computer Vision
- NLP and LLMs

Publications

- **Parisima Abdali**, Yao Wang, Li Feng, Lavanya Umaphaty (**in progress**). Gadolinium-free Contrast-enhanced MR images with Deep Learning (**Abstract submitted**).

Experience

NYU Langone Health Radiology Department | Research Scientist Intern

New York, USA

CS-MRI Reconstruction & MRI Contrast Synthesis using Deep Learning - Prof. Li Feng

July. 2023 – Present

- **Project 2:** Improved MR image generation by 55.2% through feature engineering with Constrained Contrastive Learning (CCL), enhancing the synthesis process for better medical imaging outcomes.
- Developed and utilized constraint maps using k-means clustering to classify MR tissue-specific data, aiding self-supervised learning model (CCL) in accurate feature extraction.
- Fine-tuned a U-Net model on the BraTS'21 dataset to predict T1CE images from other MR contrasts, increasing the predictive accuracy and eliminating the need for gadolinium injections, making MRI procedures easier for patients.
- Performed statistical analysis on image quality metrics (SSIM, PSNR, LPIPS), confirming statistical significance and validating the model's effectiveness.
- Implemented solutions using TensorFlow and Keras, optimizing computational processes for efficient large-scale deep learning training on high-performance computing systems.
- **Project 1:** Implemented the Magnitude Subtraction-MRI (MS-MRI) technique with the Primal-Dual Splitting Algorithm, increasing PSNR by 10.8% over Independent and 5% over TV-Temporal Compressed Sensing methods in DCE-MRI brain datasets.
- Enhanced MS-MRI by integrating parallel imaging, resulting in a 27.6% improvement in statistical metrics and image resolution, and increased robustness across multiple coil setups.
- Developed a reference-based MS-MRI technique to support 4X and 8X acceleration rates, reducing image reconstruction time and minimizing artifacts without quality loss.

NYU Courant Institute of Mathematical Sciences | Teacher Assistant

New York, USA

DL and ML Courses - Prof. Gustavo Sandoval & Prof. Chinmay Hegde

Sept. 2023 – May. 2024

- Co-led a Deep Learning Kaggle project competition using Pytorch, enriching the course with hands-on challenges.
- Mentored students in developing various deep learning projects (CV, LLMs, NLP, etc.), offering technical support, regular progress assessments, and instruction on implementing within distributed systems using HPC resources.
- Collaborated with fellow TAs in preparing lab coding exercises, enhancing the practical learning experience for students by offering diverse and programming tasks.

Imam Khomeini International University | Teacher Assistant

Qazvin, Iran

Digital Signal Processing 1 - Prof. Neda Faraji

Sept. 2021 – June. 2022

- Led weekly recitation sessions to reinforce key course concepts.
- Crafted engaging computer-based assignments in MATLAB, covering a broad spectrum of topics.

Imam Khomeini International University | Deep Learning Research Intern

Tehran, Iran

Computer Vision & Audio Processing using Deep Learning - Prof. Neda Faraji

Aug. 2020 – Dec. 2021

- Designed pipelines in MATLAB for audio data analysis, generating labels for machine learning classification tasks and fine-tuned a CNN model, achieving 75% accuracy in detecting multiple speakers on TIMIT Dataset.
- Customized and integrated advanced neural networks model (YOLOv3, MonoDepth) for real-time applications, training on the NYU Depth Dataset, which led to achieving 87.3% accuracy in social distancing estimation tasks.
- Achieved 77.5% accuracy detecting crowded areas on live video data.

Imam Khomeini International University | Teacher Assistant

Qazvin, Iran

Digital Systems Design - Prof. Seied Ali Hosseini

Dec. 2019 – June. 2020

- Collaborated with faculty and fellow TAs in weekly meetings to enhance the curriculum.
- Designed and developed written assignments and computer-based tasks using Verilog programming language.
- Led weekly recitation sessions to demystify complex topics, employing interactive teaching methods that fostered deeper understanding and engagement among students.

Selected Projects

U.S. STEM Occupations Analysis | Tableau (Portfolio) 2023

- Conducted comprehensive data cleaning and preprocessing using Tableau Prep Builder and Pandas, applying statistical and advanced data analysis techniques to prepare the dataset.
- Employed advanced data visualization and presentation skills to develop interactive Tableau dashboards on STEM employment trends, effectively conveying insights and facilitating data-driven decisions among stakeholders.

TV-Denoising and Compressed Sensing for MRI | Python (Website) 2023

- Implemented the ROF-Total Variation denoising model using the ADMM algorithm for grayscale images, achieving high-quality denoising with reduced noise. Tested different lambda levels to evaluate performance across varying Gaussian noise intensities.
- Applied the primal-dual splitting algorithm for compressed sensing on MRI data, evaluating equidistant and random 4-fold accelerated masks.

PokéGAN | Python (Github) 2023

- Designed a custom-tuned deep learning model, Generative Adversarial Network (GANs), using PyTorch to generate novel Pokémon images, incorporating data augmentation for improved synthesis quality.
- Enhanced generative model performance by integrating Autoencoders, achieving a 41.3% improvement in high-dimensional pattern recognition over the baseline.
- Optimized and accelerated computational resources for large deep learning models in distributed systems using HPC and multi-GPU clusters, achieving a 68% increase in data processing efficiency.

Modified ResNet | Python (Github) 2023

- Developed three ResNet architectures with less than 5 million trainable parameters, significantly enhancing computational efficiency and offering valuable insights for resource-constrained environments.
- Implemented techniques such as active learning rate adjustment, efficient optimizers, layer augmentation, and dropouts, achieving 94.16% training accuracy and 89.66% testing accuracy on the CIFAR-10 dataset.

VoxAlter: Real-Time Vocal Modulation | Python (Github) 2023

- Implemented real-time phase-vocoder effects, including Robotization, Whisperization, and Pitch Shifting, using STFT and ISTFT techniques for frequency-domain transformations.
- Developed a user-friendly GUI with Tkinter to provide easy access to these effects, featuring real-time plotting and demonstrating the audio transformations interactively.

Ultra-Sonic Sensor Controlled Robot | Arduino, MATLAB 2020

- Programmed an autonomous robot to navigate assigned paths using an ultra-sonic sensor and optimized its performance with a PID controller.
- Simulated the robot's system using MATLAB Simulink Control System Toolbox to analyze and improve navigation accuracy and stability.

Honors and Awards

- NYU Merit Student Scholarship (2022 - 2024)
- Graduated as the top student from Imam Khomeini International University of Iran (**Ranked 1st**) (2020)
- Winner of the "Tuition Waiver Award" from National Ministry of Science for Imam Khomeini International University, Iran (2016)
- Ranked within the top 3% among approximately 163,000 participants in the National Entrance Examination from Iranian universities, Iran (2016)

Technical Skills and Interests

Languages: Python, C/C++, MATLAB, VHDL, CUDA, Shell, LATEX, SQL

Libraries: TensorFlow, PyTorch, Pandas, NumPy, Matplotlib, Seaborn, SciPy, OpenCV, Scikit-learn, Keras

Cloud/Database: Containers, High-Performance Computing (HPC), MySQL

Tools: VScode, Git, Github, Tableau

Relevant Coursework: Probability & Stochastic Processes, Machine Learning, Deep Learning, Image and Video Processing, DSP, Medical Imaging